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System and Associated Methods for an Aerial Device to Provide Airborne Payloads

Abstract

An aerial device is provided that comprises a body member and a retention member that may include an interior space and a rearward portion that may be operable to matingly engage with a forward portion of the body member to carry a payload within the interior space, which may be in cooperation with the forward portion of the body member. The retention member may include a collar that may have an inner perimeter that may define an opening in communication with the interior space. A forward portion of the payload may extend through the opening while the rearward portion of the retention member is matingly engaged with the forward portion of the body member. The inner perimeter of the collar may substantially abut a portion of an outer surface of the payload while the retention member is matingly engaged with the body member.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to systems and methods for carrying payloads and aerial transportation and delivery thereof.

BACKGROUND OF THE INVENTION

[0002] The prior art often faces significant limitations when it comes to effective transportation of various payloads. Current solutions predominantly focus on integrating payloads with the transportation and delivery devices of the prior art. Such delivery devices and payloads of the prior art are designed and specifically intended from conception to final assembly to be self-propelled and integrated with one another. As such, the prior art provides overly complex, expensive, inefficient payload delivery systems that are often and quickly in scarce supply due to their long and complex production processes, and inconveniently intricate designs. As such, there is a need for more versatile, cost effective, simple, efficient, and/or effective solutions for non-manual aerial transportation and aerial delivery of payloads that can provide similar and comparable results with low complexity, greater economic efficiency, and greater scalability to supply demands therefor. There is also a need for payload delivery systems which can selectively carry different payload types for aerial transportation and delivery, including such payload types in large supply, and specifically including such payloads that were not intended for self-propelled aerial transportation, which generally may be efficiently mass produced compared to the prior art.

[0003] This background information is provided to reveal information believed by the applicant to be of possible relevance to the present invention. No admission is necessarily intended, nor should be construed, that any of the preceding information constitutes prior art against the present invention.

SUMMARY OF THE INVENTION

[0004] With the above in mind, embodiments of the present invention are related to an aerial device that comprises a body member and a retention member. The retention member may include an interior space and a rearward portion that may be operable to matingly engage with a forward portion of the body member to carry a payload within the interior space, which may be in cooperation with the forward portion of the body member. The retention member may include a collar that may have an inner perimeter that may define an opening that may be in communication with the interior space. A forward portion of the payload may extend through the opening while the rearward portion of the retention member is matingly engaged with the forward portion of the body member. The inner perimeter of the collar may substantially abut a portion of an outer surface of the payload while the rearward portion of the retention member is matingly engaged with the forward portion of the body member.

[0005] In some embodiments of the present invention, the retention member may include one or more of a rail that may have an end portion thereof connected to the collar. At least a portion of an inward surface of the one or more rail(s) may significantly abut a portion of the outer surface of the payload while the rearward portion of the retention member is matingly engaged with the forward portion of the body member. Some embodiments of the present invention may include one or more of an engagement member that may be mounted on the body member and/or the retention member. The one or more engagement member may be operable to be reversibly engaged with an engagement unit of an aerial vehicle.

[0006] Some embodiments of the present invention may include a thruster that may be to selectively provide a thrust force, and one or more fins that may be connected to and extending from the body member. At least a portion of each one of the fins may be operable to rotatably move. Some embodiments of the present invention may further include wings that may be connected to the body member and/or the retention member. At least a portion of each of the wings may be operable to rotatably move.

[0007] Some embodiments of the present invention may include a travel propulsion member that may be mounted on and extending from a portion of the body member and/or a portion of the retention member. The travel propulsion member may be operable to generate a travel thrust in a rearward direction.

[0008] Some embodiments of the present invention may further include one or more of a controller, a sensor, and a power unit. The controller may be in communication with the thruster and the fins. The sensor(s) may be positionable in communication with the controller and may be operable to sense one or more of a characteristic data point. The power unit may be in communication with the controller and/or the at least one sensor. The sensor(s) may be is operable to provide a sensor signal based on the characteristics data point. The controller may be operable to selectively control the thruster and/or the fins based on the sensor signal.

[0009] Some embodiments of the present invention may include one or more of a body communication point and a retention communication point. The body communication point(s) may be carried by the body member, and the retention communication point(s) may be carried by the retention member. The controller and/or the power unit may be carried by the body member and may be in communication with one or more of the body communication point(s). One or more of the sensor(s) may be carried by the retention member and may be in communication with one or more of the retention communication point(s).

[0010] One or more of the body communication point(s) may be engaged in communication with one or more of the retention communication point(s) when and/or while the rearward portion of the retention member is matingly engaged with the forward portion of the body member. In some embodiments of the present invention, the controller may be operable to determine one or more of a path of travel based on the sensor signal. The controller may be further operable to selectively control one or more of the thruster and/or the fins based on one or more of the path(s) of travel determined.

[0011] Some embodiments of the present invention may be directed to an aerial device comprising a body member, a retention member, one or more engagement members, a thruster, and one or more fins. The retention member may include an interior space and a rearward portion that may be operable to matingly engage with a forward portion of the body member to carry a payload within the interior space, which may be in cooperation with the forward portion of the body member. The one or more engagement member(s) may be mounted on the body member and/or the retention member. The thruster may be operable to selectively provide a thrust force. The fins may be connected to and/or extending from the body member.

[0012] The retention member may include a collar that may have an inner perimeter that may define an opening that may be in communication with the interior space. A forward portion of the payload may extend through the opening when and/or while the rearward portion of the retention member is matingly engaged with the forward portion of the body member. The inner perimeter of the collar may substantially abut a portion of an outer surface of the payload when and/or while the rearward portion of the retention member is matingly engaged with the forward portion of the body member.

[0013] The retention member may include one or more rails that may have an end portion thereof connected to the collar. At least a portion of an inward surface of the one or more rails may significantly abut a portion of the outer surface of the payload when and/or while the rearward

portion of the retention member is matingly engaged with the forward portion of the body member. The one or more engagement member(s) may be operable to be reversibly engaged with an engagement unit of an aerial vehicle.
[0014] Some embodiments of the present invention may be directed to an aerial device that comprises a body member, a retention member, one or more engagement member(s), a thruster, fins, a controller, one or more sensors, a power unit, body communication point(s), and retention communication point(s). The retention member may include a collar and one or more rails. The rail(s) may include a forward end portion connected to the collar. The engagement member(s) may be mounted on the body member and/or one or more of the rail(s). The thruster may be operable to selectively provide a thrust force. The fins may be connected to and/or extending from the body member.
[0015] The controller in communication with the thruster and/or the fins. The sensor(s) may be operable to sense at least one characteristic data point. The power unit may be in communication with the controller, and the power unit may be positionable in communication with one or more of the sensor(s). The body communication point(s) may be carried by the body member. The retention communication point(s) may be carried by one or more of the rail(s).
[0016] The collar and the rail(s) may define an interior space. The collar may have an inner perimeter that may define an opening that may be in communication with the interior space. The rail(s) may include a rearward end portion that may be operable to matingly engage with a forward portion of the body member to carry a payload within the interior space, which may be in cooperation with the forward portion of the body member. A forward portion of the payload may extend through the opening when and/or while the rearward portion the one or more rail(s) is matingly engaged with the forward portion of the body member.
[0017] The inner perimeter of the collar and a portion of an inward surface of the rail(s) may substantially about a portion of an outer surface of the payload when and/or while the rearward portion of the rail(s) is matingly engaged with the forward portion of the body member. The engagement member(s) may be operable to be reversibly engaged with an engagement unit of an aerial vehicle. The sensor(s) may be operable to provide a sensor signal based on the at least one characteristics data point. A portion or more of each one of the fins may be operable to rotatably move.
[0018] The controller may be operable to selectively control one or more of the thruster and the fins based on the sensor signal. The controller and the power unit may be carried by the body member and may be in communication with one or more of the body communication point(s). One or more of the sensor(s) may be carried by the retention member and may be in communication with one or more of the retention communication point(s). One or more of the body communication point(s) may engage in communication with one or more of the retention communication point(s) when and/or while the rearward portion of the rail(s) is matingly engaged with the forward portion of the body member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0019] Some embodiments of the present invention are illustrated as an example and are not limited by the figures of the accompanying drawings, in which like references may indicate similar elements.
[0020] FIG. **1** is a schematic illustration of an aerial device according to an embodiment of the present invention.
[0021] FIG. **2** is another schematic illustration of the aerial device according to FIG. **1**, showing an air brake in the deployed position.

[0022] FIG. **3** is a schematic illustration of the aerial device according to FIG. **1**, shown with wings.
[0023] FIG. **4** is a schematic illustration of the aerial device according to FIG. **3**, shown with wings and a travel propulsion member.
[0024] FIG. **5** is a schematic illustration and partially exploded view of the aerial device according to FIG. **1**.
[0025] FIG. **6** is a schematic illustration of the aerial device according to FIG. **1**, with the aerial device illustratively shown being deployed from an aerial vehicle.

DETAILED DESCRIPTION OF THE INVENTION

[0026] The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. Those of ordinary skill in the art realize that the following descriptions of the embodiments of the present invention are illustrative and are not intended to be limiting in any way. Other embodiments of the present invention will readily suggest themselves to such skilled persons having the benefit of this disclosure. Like numbers refer to like elements throughout.
[0027] Although the following detailed description contains many specifics for the purposes of illustration, anyone of ordinary skill in the art will appreciate that many variations and alterations to the following details are within the scope of the invention. Accordingly, the following embodiments of the invention are set forth without any loss of generality to, and without imposing limitations upon, the claimed invention.
[0028] In this detailed description of the present invention, a person skilled in the art should note that directional terms, such as “above,” “below,” “upper,” “lower,” and other like terms are used for the convenience of the reader in reference to the drawings. Also, a person skilled in the art should notice this description may contain other terminology to convey position, orientation, and direction without departing from the principles of the present invention.
[0029] Furthermore, in this detailed description, a person skilled in the art should note that quantitative qualifying terms such as “generally,” “substantially,” “mostly,” and other terms are used, in general, to mean that the referred to object, characteristic, or quality constitutes a majority of the subject of the reference. The meaning of any of these terms is dependent upon the context within which it is used, and the meaning may be expressly modified.
[0030] An embodiment of the invention, as shown and described by the various figures and accompanying text, provides an aerial device **100** to house and deliver a variety of payloads **102**. Those skilled in the art may notice and appreciate that the various embodiments of the aerial device **100** as described herein may advantageously provide for the housing and delivery of payloads **102** comprising a variety of different shapes, sizes, and compositions, and specifically, for the housing and delivery of payloads **102** that normally were not, and/or could not be, housed and delivered by previous other aerial devices. More specifically, the aerial device **100** may be self-propelled and deployable from aircraft or other means of deployment, such as an aerial vehicle **602**. The aerial device **100** may be modular and/or shaped and sized to house and deliver a variety of payloads **102** which comprise a variety of shapes, sizes, and compositions and that generally were not housed and/or delivered by self-propelled aerial devices for one reason. With the reasons including to lack of planning, unintended use, unintended purpose, failure to achieve such use or implementation, and/or unforeseen use and/or unforeseen implementation.
[0031] Initially referring to FIG. **1**, the aerial device **100** may comprise a body member **110**, a retention member **104**, a thruster **116**, and one or more fins **112**. The aerial device **100** may be sized and configured to house, carry, and/or retain a payload **102**. The retention member **104** may act in cooperation with the body member **110** to house, carry, and/or retain a payload **102**. At least a

portion of the retention member **104** may be configured, adapted, and/or operable to be removably, and/or non-removably, engaged and/or matingly engaged, with the body member **110** and/or with a forward portion of the body member **110**, which may be to house, carry, secure, and/or retain the payload **102**, and/or to house, carry, secure, and/or retain the payload **102** in cooperation with the body member **110** and/or the forward portion of the body member **110**. All of, or at least a portion of the payload **102** may be positioned within an interior area/space of the aerial device **100** and/or an interior area/space the retention member **104** when and/or while the payload **102** is being housed, carried, and/or retained by the aerial device **100** and/or the retention member **104**.
[0032] The thruster **116** may selectively provide a thrust force to cause the aerial device **100** and the payload **102** to be moved in a direction opposite from, and/or inverse of, the thrust force provided by the thruster **116**. The thruster **116** may be positioned at an end of the aerial device **100** and/or the body member **110** that is opposite the retention member **104** and/or the payload **102** carried by the retention member **104** and/or by the aerial device **100**. The thrust force provided by the thruster **116** may cause an exhaust **114** and/or propellant **120** to be released from the thruster **116** in the direction of the thrust force, which may be in a direction opposite and/or inverse to the direction of movement of the aerial device **100** and the payload **102** caused by the thrust force generated by the thruster **116**.
[0033] The fins **122** may be positioned extending outwardly from one or more portions of the aerial device **100**, the body member **110**, and/or the retention member **104**. The fins **112** may be utilized to provide for aerial steering, guidance, and/or control over a flight trajectory of the aerial device **100**. The fins **112** may be controllable to move and change the position and angle of the fin **112** relative to the body member **110** and/or relative to the aerial device **100** to steer, guide, and control the flight trajectory of the aerial device **100**.
[0034] Embodiments of the present invention may also include one or more engagement members **118** which may be mounted on a portion of an outer surface of the body member **110**, the retention member **104**, and/or the rail(s) **108**. The engagement members **118** may be operable to be removably engaged and/or removably matingly engaged with other objects and components, such as, but without limitation, with a respective engagement unit **604** of an aerial vehicle **602**. More details on the engagement members **118** follows further below.
[0035] Now additionally referring to FIG. **2**, embodiments of the present invention may include an air brake **203**. The air brake **203** may be carried by the body member **110**, and the air brake **203** may be utilized to slow the velocity of the aerial device **100** when the aerial device **100** is in flight. The air brake **203** may be deployable, and controllable to be deployed, from the aerial device **100**. The air brake **203** may comprise one or more of a drogue chute, drag chute, and/or a parachute. However, it is contemplated that the air brake **203** may comprise other types of air brakes such as hinged spoiler brake or speed brake panel hingedly mounted on the body member **110** or the fins **112**. The air brake **203** may be deployable from a stored position to a deployed position. The stored position of the air brake **203** may be defined as when the air brake **203** at least mostly contained within the body member **110** and/or when the air brake **203** is not extended outwardly from the body member **110**. The deployed position of the air brake **203** may be defined as when the air brake **203** is at least partially extended outwardly from the body member **110**, such that the air brake **203** may cause an air drag force against the body member **110** and/or against the aerial device **100** to attempt to slow the air velocity of the body member **110** and/or of the aerial device **100**.
[0036] In some embodiments of the present invention, the retention member **104** may comprise one or more of a collar **106** and a rail **108**. The collar **106** may comprise a ring shape and/or a conical shape having an opening that may be defined by an inner perimeter of the collar **106**. The opening may be positioned at one or both ends of the collar **106**. The opening may be extending from and/or between the interior area/space of the aerial device **100**/of the retention member **104** and through and outwardly from the collar **106**. The opening may be in communication and/or in fluidic communication with the interior space of the retention member **104** and/or with an environment

adjacent to, external to, and/or exterior to, the collar **106** and/or the aerial device **100**. The collar **106** may be shaped to have an inner perimeter surface of the collar **106** to be significantly abutting against an outside surface of a payload **102** that is, or to be, carried by the aerial device **100**. The collar **106** may be connected with an end portion of the rail(s) **108**. The collar **106** may also be connectable, removably connectable, matingly engaged, and/or removably matingly engaged with an end portion of the rail(s) **108**.

[0037] The rails **108** may be comprise elongated extensions. The rails **108** may also comprise a single rail **108** that may have cylindrical shape and may have an opening at one or both ends of the cylindrical rail **108**. It is contemplated that embodiments of the present invention may include any number of rails **108** and of rails **108** that comprise a variety of geometric shapes as may be understood by those who may have skill in the art. Thus, for the purposes of the present description of the present invention, the use of the term “rail **108**” and “rails **108**” should be understood to be used interchangeably as embodiments of the present invention may include a rail **108**, may also include a plurality of rails **108**.

[0038] The rails **108** may be formed and/or shaped such that at least a portion of an inward facing surface/at least a portion of a lower surface of the rails **108** may significantly abut against at least a portion of an outer surface of a payload **102** when and/or while the payload **102** is carried by the aerial device **100** and/or while the rearward portion of the retention member **104** and/or rails **108** is engaged with the forward portion of the body member **110**. The rails **108** may be connected to, removably connectable with, engaged, and/or removably engaged with at one end with a portion of the collar **106**. However, is contemplated that in some embodiments of the present invention, the rail(s) **108** and the collar **106** may be connected to each other as a single monolithic piece as the retention member **104**. The rails **108** may also be connected to, removably connectable with, engaged, and/or removably engaged with a portion of the body member **110** and/or with a respective body connector **202** of the body member **110**. Also, in some embodiments of the present invention, the rails **108** may be connected to, removably connectable with, engaged, and/or removably engaged with a forward portion of the body member **110** that is at an opposing side of the body member **110** as compared to a rearward portion of the body member **110** that at which the thruster **116** may be carried by the body member **110**.

[0039] The body member **110** may comprise one or more body connectors **202**. Embodiments of the present invention may include the same number of body connectors **202** as there are rails **108** and/or retention members **104**. Each body connector **202** may be configured to be connected to, removably connectable with, engaged, and/or removably engaged with a respective rail **108**. In embodiments of the present invention that include a single rail **108** having a cylindrical shape and/or a monolithic retention member **104** having a cylindrical shape, the body connector **202** may encompass at least a portion of a forward perimeter of the body member **110**. The body connector(s) **202** may be controllable to be reversibly engaged with, and/or reversibly connected to, the one or more rails **108** and/or the retention member **104**.

[0040] Now referring to FIGS. **1-6**, some embodiments of the present invention may include a number of electronically powered and/or electronically controlled components. For example, but without limitation, the aerial device **100** may include one or more of a controller **204**, a communication unit **206**, a power unit **208**, a data store **210**, a data bus **212**, sensors **214**, and action members **216**. To increase the level of terseness, for the purposes of the description of the present invention, and without limitation, the terms “electronically powered elements/components/members/units,” “electronically controlled elements/components/members/units,” “electronically powered elements/components/members/units of the aerial device **100**, and “electronically controlled elements/components/members/units of the aerial device **100**” may be used interchangeably, and without limitation, to refer to one or more of the controller **204**, the communication unit **206**, the power unit **208**, the data store **210**, the data bus **212**, the sensors **214**, the action members **216**, the thrusters **116**, the fins **112**, the payload **102**, the engagement members **118**, the air brake **203**, wing(s) **302**, the travel member **402**, and the communication point **502** collectively, individually, and/or in any combination(s) thereof. More details on the wing(s) **302**, the travel member **402**, and the communication point **502** follows further below.

[0041] To further increase the level of terseness throughout the present application, since many components, devices, units, and members described herein may be in communication with one another, for the purposes of the present invention the terms "computer-readable communication elements/components/members/units" and "computer-readable communication elements/components/members/units of the aerial device **100**" may be used interchangeably, and without limitation, to refer to one or more of the controller **204**, the communication unit **206**, the power unit **208**, the data store **210**, the data bus **212**, the sensors **214**, the action members **216**, the thrusters **116**, the fins **112**, the payload **102**, the engagement members **118**, the air brake **203**, the wings **302**, the travel member **402**, the communication point **502**, a communication device **606**, the aerial vehicle **602**, and the engagement unit(s) **604** collectively, individually, and/or in any combination(s) thereof. More details on the communication device **606**, the aerial vehicle **602**, and the engagement unit(s) **604** follows further below.

[0042] The controller **204** may be carried by the body member **110**. The controller **204** may be in communication with one or more of the electronically controlled components of the aerial device **100** and/or with one or more of the computer-readable communication elements. The controller **204** may be utilized to monitor, manage, selectively control and/or selectively operate one or more of the electronically controlled components of the aerial device **100** and/or with one or more of the computer-readable communication elements. The controller **204** may be configured to read, write, send, receive, compute, calculate, execute, process, and/or manage computer-readable data, information, code, instructions, and/or signals. Examples of the controller **204** include, but without limitation, a processor, microprocessor, microcontroller, central processing unit, graphics processing unit, field-programmable gate array, non-field-programmable gate array, and any combination thereof and any other machine processing unit that may be used as the controller **204** as may be understood by those who may have skill in the art. More details on the controller **204** follows throughout below.

[0043] The communication unit **206** may be carried by the body member **110**. The communication unit **206** may be in communication with one or more of the electronically controlled components of the aerial device **100** and/or with one or more of the computer-readable communication elements. The communication unit **206** may be utilized to control, regulate, manage, relay, process, facilitate, and/or operate any computer-readable data, information, code, instructions, and/or signals to, from, and/or between one or more of the electronically controlled components of the aerial device **100** and/or with one or more of the computer-readable communication elements. Examples of the communication unit **206** may include, but without limitation, a network card, a transceiver, a receiver, a relay, a network switch, a router, a network modem, a hub, a radio, and antenna, and any combination thereof and any other device that may be used as the communication unit **206** for computer-readable communications as may be understood by those who may have skill in the art. More details on the communication device **606** and the aerial vehicle **602** follows further below.

[0044] The power unit **208** may be carried by the body member **110**. The power unit **206** may be in communication with one or more of the electronically controlled components of the aerial device **100**. The power unit **208** may be utilized to provide power, regulate power, monitor power, and/or control power, to and/or from one or more of the electronically controlled components of the aerial device **100**. The power unit **208** may comprise one or more of, but without limitation, a fuel cell, a battery, a rechargeable battery, a generator, a capacitor, a solar cell, a power store, and any combination thereof and any other power providing device to use as the power unit **208** to provide power, such as electrical power, as may be understood by those who may have skill in the art.

[0045] In some embodiments of the present invention, the power unit **208** may include a fuel storage. The fuel storage of the power unit **208** may be in fluid communication with the thrusters **116**, and the fuel storage of the power unit **208** may be utilized to provide fuel to the thrusters **116**. The thrusters **116** may receive the fuel from the fuel storage of the power unit **208** and the thrusters **116** may utilize and combust the fuel to generate the thrust/thruster force. In other embodiments of the present invention, the thrusters **116** may include its own fuel storage to be utilized and combusted by the thrusters **116** to generate the thrust/thruster force.

[0046] The data store **210** may be carried by the body member **110**. The data store **210** may be in communication with one or more of the electronically controlled components of the aerial device **100**, and/or the data store **210** may also be in communication with one or more of the computer-readable communication elements. The data store **210** may include a computer-readable memory unit. The data store **210** may be utilized to read, write, store, send, receive, index, compute, and/or manage computer-readable data, information, code, instructions, and/or signals to and/or from one or more of the computer-readable communication elements. Examples of the data store **210** includes, but without limitation, a hard drive, a disk drive, magnetic tape, a solid-state drive, a floppy disk, a hard disk, computer-readable datastore, computer-readable memory, random access memory, non-random access memory, volatile computer-readable memory, non-volatile computer-readable memory, and any combination thereof and any other data storage device that may be used as the data store **210** as described herein as may be understood by those who may have skill in the art.

[0047] The data bus **212** may be carried by the body member **110**. The data bus **212** may be in communication with one or more of the electronically controlled components of the aerial device **100**. The data bus **212** may be utilized to control, regulate, manage, relay, process, facilitate, and/or operate any computer-readable data, information, code, instructions, and/or signals to, from, and/or between one or more of the electronically controlled components of the aerial device **100**. Examples of the data bus **212** includes, but without limitation, a motherboard, a printed circuit board, a break-out board, a wiring harness, a communication relay, and any other components that may be utilized as the data bus **212** to facilitate computer-readable communication as may be understood by those who may have skill in the art.

[0048] Now referring to FIGS. **2-6**, some embodiments of the aerial device **100** may include one or more action members **216**. The action member **216** may be carried by the body member **110**, and the action members **216** may be utilized to selectively cause one or more of the electronically controlled members of the aerial device **100** to move, actuate, and/or rotate relative to the action member **216** and/or relative to the body member **110**. The action members **216** may also be utilized to cause one or more of the retention member **104** and/or the rail(s) **108** to be engaged with, connected to, removably engaged with, and/or removably connected to the body member **110**, the forward portion(s) of the body member **110**, the forward portion(s) of the body member **110**, and/or the body connector(s) **202**. The action member(s) **216** may also be utilized to cause the fin(s) **112** to move and/or change the position and angle of the fin(s) **112** relative to the body member **110** and/or relative to the aerial device **100**, to steer, guide, and/or control the flight trajectory of the aerial device **100**. The action member(s) **216** may also be utilized to actuate, deploy, release, and/or activate the air brake(s) **203** to cause the air brake(s) **203** to slow the velocity of the aerial device **100** when the aerial device **100** is in flight. The action member(s) **216** may also be utilized to grasp, retain, and/or be removably engaged with the payload **102** when the payload **102** is being carried by the body member **110**, the retention member **104**, the collar **106**, and/or the rail(s) **108**. The controller **204** may selectively control and/or selectively operate the functions, operations, and/or features of the action member(s) **216**.

[0049] Some embodiments of the aerial device **100** may include one or more sensors **214**. The sensors **214** may be mounted on and/or carried by one or more of the body member **110**, by the retention member **104**, and/or by the collar **106**. The sensors **214** may be in communication with one or more of the electronically controlled components of the aerial device **100**. The sensors **214** may be utilized to sense and/or detect one or more characteristic data points. For example, but without limitation, the sensors **214** may be utilized to sense and/or detect characteristic data points including, but without limitation, one or more of the relative location of the sensor **214**, the body member **110**, the retention member **104**, the collar **106**, and/or of the aerial device **100**, the temperature of the local environment and/or the temperature of the payload **102** and/or aerial device **100**, the relative location of the aerial device **100** with respect to an object or surface, ground speed of the aerial device **100**, wind speed of the aerial device **100**, acceleration/deceleration of the aerial device **100**, projected trajectory of the aerial device **100**, fuel level and/or energy charge of the power unit **208**, g-forces on the sensor(s) **214**, the aerial device **100**, and/or on the payload **102**, the humidity level of the local environment, the current status of one or more of the electronically controlled components of the aerial device **100**, the location of an object or target **608**, and/or the projected path of travel of an object or target.

[0050] The sensor(s) **214** may emit a sensor signal related to the characteristic data point(s) detected and/or sensed by the sensor(s) **214**, which may be received by the controller **204**. The controller **204** may cause one or more predetermined actions to take place based an identification signal received by the controller **204**. The controller **204** may also be configured to selectively control and cause one or more of the electronically controlled components of the aerial device **100** to take a predetermined action based one an identification signal received by the controller **204**. Examples of the sensor(s) **214** may include, but without limitation, one or more of a temperature sensor, infrared sensor, ultraviolet light sensor, optical sensor, spatial sensor, proximity sensor, laser imaging sensor, radar sensor, ultrasonic sensor, x-ray sensor, visible light sensor, humidity sensor, air pressure sensor, global positioning satellite transceiver, magnetometer, vibration sensor, load sensor, pitot-static sensor, angle of attack sensor, oxygen sensor, ice detector, collision avoidance sensor, terrain sensor, fire sensor, smoke sensor, doppler lidar sensor, anemometer, doppler radar sensor, accelerometer, gyroscope, tachometer, pitot tube, wind speed sensor, and ground speed sensor. More details on the sensors **214**, the controller **204**, and the identification signals follows further below.

[0051] Now specifically referring to FIG. **6**, embodiments of the aerial device **100** may be configured to be reversibly attached to an aerial vehicle **602**. The aerial vehicle **602** may comprise, but without limitation, an airplane, helicopter, glider, unmanned aerial vehicle, and/or an ultralightweight flying craft. The aerial device **100** may be reversibly attached to the aerial vehicle **602** via one or more of the engagement members **118** of the aerial vehicle **602** being removably engaged with a respective engagement unit **604** of the aerial vehicle **602**. One or more of the electronic components of the aerial device **100**, such as the controller **204** and/or the communication unit **206**, may be positioned in communication with the aerial vehicle **602** when/while one or more of the engagement members **118** are engaged with an engagement unit **604** of the aerial vehicle **602**, such that the communication is via the engagement members **118** and their respective engaged engagement units **604**. However, it is contemplated that one or more of the communication components and/or electronically controlled components of the aerial device **100** may be in wireless communication with the aerial vehicle **602**, which may be communication via a network/communication transceiver **606**. More details on the network/communication transceiver **606** follows further below.

[0052] In some embodiments of the present invention, the engagement members **118** of the aerial device **100** may be configured to be removably/reversibly attached to, and/or engaged with, preexisting engagement units **604** on current aerial vehicles **602**. Those skilled in the art may notice and appreciate that with the engagement members **118** being configured to be removably/reversibly attached to, and/or engaged with, preexisting engagement units **604** on current aerial vehicles **602** that no modification an aerial vehicle **602** may be required to allow the engagement members **118** of the aerial device **100** to be removably/reversibly attached to, and/or engaged with, the engagement units **604** of the aerial vehicle **602**. For example, but without limitation, in some embodiments of the present invention the engagement members **118** may be configured to be removably/reversibly attached to, and/or engaged with, engagement units **604** of aerial vehicles **602** that comprise one or more of a hardpoint connection, a pylon connection, an ejector release unit, a rack connector, a munition launcher connector, a fuel tank connector, a munition station connector, a munition pod connector, drop tank connector, and any combinations thereof and/or any other external connector on an aerial vehicle **602** as may be understood by those who may be skilled in the art.

[0053] The aerial vehicle **602** may be configured to control, monitor, and/or otherwise manage the aerial device **100** that the aerial vehicle **602** is in communication with to control one or more of the operations of the electronically controlled components of the aerial device **100** and/or monitor the status of one or more of the communication components and/or electronically controlled components of the aerial device **100**. For example, but without limitation, the status monitored of the aerial device **100** by the aerial vehicle **602** may include one or more of the weight of the aerial device **100**, the characteristics of the payload **102** carried by the aerial device **100**, the fuel and/or energy capacity of the power unit **208**, and/or the estimated maximum flight range and estimated maximum flight time of the aerial device **100**. The aerial device **602** may communicate and control the aerial device **100** to follow a particular path of travel **610***a-e* for a particular target **608**.

[0054] The aerial vehicle **602** may also communication a target **608** to the aerial device **100** and one or more of the electronic components of the aerial device **100**, such as the controller **204**, may determine a path of travel **610***a-e* for the aerial device **100** to follow upon the aerial device **100** being released from the aerial vehicle **602**. The one or more the electronic components of the aerial device **100**, such as the controller **204**, may also determine a path of travel **610***a-e* for the aerial device **100** to follow upon the aerial device **100** being released from the aerial vehicle **602** based upon one or more flight factors. The flight factors may include one or more of the current weather conditions, one or more characteristics of the payload **102** carried by the aerial device **100**, and/or a path of travel signal received by one or more of the electronic components of the aerial device **100** from the aerial vehicle **602**. The characteristics of the payload **102** may include one or more of the weight of the payload **102**, the mass of the payload **102**, the fragility of the payload **102**, the volatility of the payload **102**, the maximum acceleration and/or g-force the payload **102** may be subjected to, and/or the type of payload **102**.

[0055] Embodiments of the aerial device **100** may be configured to carry and deliver a payload **102** to a target **608**. The payload **102** carried by the aerial device **100** may include, for example, but without limitation, one or more of food, water, consumables, medical supplies, equipment, shells, clothing, materials, and/or chemical substances, compounds, or mixtures thereof. The electronic components of the aerial device **100**, such as the controller **204** based on the sensor signal from the sensor(s) **214**, may be configured to determine and identify a path of travel **610***a-e* for the aerial device **100** to follow from the current position of the aerial device **100** for the aerial device **100** to propel and navigate itself to the target **608** and/or for the aerial device **100** to deliver the payload **102** carried by the aerial device **100** to the target **608**. For example, but without limitation, the controller **204** may determine and identify a path of travel **610***a-c* based on the sensor signal from the sensor(s) **214** for the aerial device **100** to navigate to follow for the aerial device **100** to travel by selectively operating, commanding, and/or controlling the thruster **116**, the fins **112**, the wings **302**, the travel propulsion member **402**, and/or the air brake **203** to arrive at the target **608** to deliver the payload **102**.

[0056] Another example, but without limitation, the controller **204** may determine and identify a path of travel **610***d-e* based on the sensor signal from the sensor(s) **214** for the aerial device **100** to navigate to follow for the aerial device **100** to travel and perform a predetermined action at an action point **612** along the path of travel **610***d-e* in order for the aerial device **100** to deliver the payload **102** to/at the target **608**. For example, upon the aerial device **100** arriving at one of the action points **612** along a path of travel **610***d-e* and/or the controller **204** determining that the aerial device **100** is at an action point **612** along a path of travel **610***d-e* based on the sensor signal from the sensor(s) **214**, the controller **204** may cause one or more of the electronic components of the aerial device **100** to take a predetermined action including one or more of: moving the air brake **203** to the deployed position from the stored position; selectively controlling the thruster(s) **116** to change the amount of and/or stop generating a thrust force; changing the orientation of one or more of the fins **112**; activating the payload **102**, such as commencing a chemical reaction by the payload **102**; and, disengaging the retention member **104**, collar **106**, rail(s) **108**, body member **110**, body connector **202**, and/or action members **216** from one another such as to release the retention member **104** and/or the payload **102** from the aerial device **100** and/or from the body member **110** to allow for the payload **102** to be delivered to the target **608**. In some embodiments of the present invention, the payload **102** may include a parachute and/or its own air brake that may deploy upon the payload **102** being released from the aerial device **100** to slow the fall of the payload **102** to the target **608**.

[0057] One or more of the electronic components of the aerial device **100**, such as the controller **204** in cooperation with the sensor(s) **214** via the sensor signal from the sensor(s) **214**, may be configured to detect and determine at what point along a path of travel **610***a-e* that the aerial device **100** has arrived at an action point **612**. For example, but without limitation, the controller **204** in cooperation with the sensor(s) **214** and the sensor signal from the sensor(s) **214** may detect and determine that the aerial device **100** has arrived at an action point **612** upon the controller **204** determining that the sensor signal from the sensor(s) **214** indicates an action point trigger. The controller **204** may determine that an action point trigger is indicated by the sensor signal when the sensor signal includes data facts that in summation indicate that the aerial device **100** is at an action point **612** including, but without limitation, current proximity with a target **608**, current wind speed, current ground speed, current altitude, current location relative to the ground surface/terrain **614**, current distance from the ground surface/terrain **614**, and current time of day.

[0058] In some embodiments of the present invention, one or more of the electronic communication components of the aerial device **100** may be in communication with a network/communication transceiver **606**. For the purposes of the present invention, the network/communication transceiver **606** may be referred to as a communication transceiver **606** without any limitation. The communication transceiver **606** may comprise a communication network and/or a communication device that is in communication with the communication network. The communication network may comprise any network utilized to monitor, manage, facilitate, and/or allow computer-readable communications to be sent thereto, therefrom, and therewith. Examples of the communication network include, but without limitation, a local access network, satellite telecommunications, a wireless communication network, a cell tower communication network, a wide-access network, blue-tooth communications, microwave communications, x-ray communications, and/or at least one network communication server and any combinations thereof, and/or any other network and communication network type as may be understood by those who may have skill in the art.

[0059] The communication device may be in communication with one or more of the electronically controlled and/or electronic communication components of the aerial device **100** via the communication network. The communication device may be utilized to monitor the status of, and/or to control the functions of, one or more of the electronically controlled components of the aerial device **100**. Examples of the communication device may include, but without limitation, one or more of a computer machine, base station, portable user device, and/or any combinations thereof.

[0060] Now referring to FIG. **3**, embodiments of the present invention may include one or more wings **302**. The wings **302** may be connected to and/or moveably attached to the body member **110**, the retention member **104**, the rail(s) **108**, and/or to the collar **106**. The wings **302** may be stationary in relation to the body member **110** of the aerial device **100**, and in some embodiments of the present invention, at least a portion of the wings **302** may be rotatably movable about an axis extending through at least a portion of the wing **302**, the aerial device **100**, the retention member **104** and/or the body member **110**body member **110**. In some embodiments of the present invention, the wings **302** may include a rotatable portion thereof that may hingedly rotate in relation to its respective wing **302**. The wings **302** may be in communication with, and/or selectively controllable by, one or more of the electronic components of the aerial device **100**. Each wing **302** may include an action member **216** that may be selectively controlled by the controller **204** to control and cause the wing **302**, and/or a rotatable portion of the wing **302** to rotatably move. The rotational movement of the wings **302** and/or of the rotatable portion of the wings **302** may be utilized to control the flight and/or aerial navigation of the aerial device **100** when the aerial device **100** is flying through the air by the trust force generated by the thruster(s) **116**.

[0061] Now referring to FIG. **4**, in some embodiments of the present invention, the aerial device may include a travel propulsion member **402**. The travel propulsion member **402** may be mounted on and extending from a portion of the body member **110**, the retention member **104**, and/or the rail(s) **108**. The travel propulsion member **402** may be in communication with, and/or selectively controllable by, one or more of the electronic components of the aerial device **100**, such as with/by the controller **204** and with the power unit **208**. The travel propulsion member **402** may be utilized to selectively generate a travel thrust force in a rearward direction that may be substantially parallel and aligned with a central and/or longitudinal axis of the aerial device **100**, the body member **110**, and/or the retention member **104**body member **110**body member **110**. Examples of the travel propulsion member **402** include, but without limitation, a jet engine, a combustion engine, a propeller, a rotor, an in-line engine, an electric motor, and any combinations thereof. Those skilled in the art may notice and appreciate that the travel propulsion member **402** may provide a more fuel-efficient source of thrusting forces to allow embodiments of the aerial device **100** to travel for a longer period of time than embodiments of the aerial device **100** that only include thruster(s) **116**.

[0062] Now referring to FIGS. **5-6**, some embodiments of the aerial device **100** may include communication points **502**. The communication points **502** may be positioned at a portion of one or more of the retention member **104**, the rail(s) **108**, the body member **110**, and/or the body connector(s) **202**. The communication point **502** may be positioned such that at least a pair of the communication point **502** come into communicative contact with one another when the retention member **104** and/or rail(s) **108** are removably/reversibly engaged with, connected to, and/or attached to the body member **110** and/or the body connector **202** of the body member **110** so as to allow communication between one or more of the electronic components of the aerial device **100** that may be separately connected to, and/or carried by, the body member **110** and the retention member **104** and/or rail(s) **108**.

[0063] Now referring to FIG. **6**, for the purposes of the present invention, the target **608** may include a verity of objects, surfaces, locations, and positions. For example, but without limitation, the target **608** may comprise one or more of a location on the ground surface/terrain **614**, a structure, a building, a vehicle, a means of transportation, a marked point, a supply drop point, a delivery point, a landing pad, a runway, a drop zone, an area of the ground surface/terrain **614**, and/or any combinations thereof and/or any other object, surface, location, or position as may be understood by those who may have skill in the art. The sensor(s) **214** may be configured to sense one or more characteristic data point(s) pertaining to a target **608** and relay the sensed characteristic data point(s) to one or more of the electronic components of the aerial device **100**, such as to the controller **204**, as data sensor signal. For example, but without limitation, the sensor(s) **214** may be configured to sense information pertaining to a target **608** including one or more of, a target type, a target location, a target projected movement path, a target composition, relative distance to the target in relation to the aerial device **100**, target temperature, and/or any combinations thereof.

[0064] Some of the illustrative aspects of the present invention may be advantageous in solving the problems herein described and other problems not discussed which are discoverable by a skilled artisan.

[0065] While the above description contains much specificity, these should not be construed as limitations on the scope of any embodiment, but as exemplifications of the presented embodiments thereof. Many other ramifications and variations are possible within the teachings of the various embodiments. While the invention has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made, and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention is not limited to the particular embodiment disclosed as the best or only mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the description of the invention. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited. Moreover, the use of the terms first, second, etc. do not denote any order or importance, but rather the terms first, second, etc. are used to distinguish one element from another. Furthermore, the use of the terms a, an, etc. do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item.

Claims

1. An aerial device comprising: a body member; and a retention member having an interior space and a rearward portion that is operable to matingly engage with a forward portion of the body member to carry a payload within the interior space in cooperation with the forward portion of the body member; wherein the retention member includes a collar having an inner perimeter that defines an opening that is in communication with the interior space; wherein a forward portion of the payload extends through the opening while the rearward portion of the retention member is matingly engaged with the forward portion of the body member; and wherein the inner perimeter of the collar substantially abuts a portion of an outer surface of the payload while the rearward portion of the retention member is matingly engaged with the forward portion of the body member.

2. The aerial device according to claim 1, wherein the retention member includes at least one rail having an end portion connected to the collar.

3. The aerial device according to claim 2, wherein at least a portion of an inward surface of the at least one rail significantly abuts a portion of the outer surface of the payload while the rearward portion of the retention member is matingly engaged with the forward portion of the body member.

4. The aerial device according to claim 1, further comprising at least one engagement member mounted on at least one of the body member and the retention member; wherein the at least one engagement member is operable to be reversibly engaged with an engagement unit of an aerial vehicle.

5. The aerial device according to claim 1, further comprising a thruster to selectively provide a thrust force; and a plurality of fins connected to and extending from the body member; wherein at least a portion of each one of the plurality of fins is operable to rotatably move.

6. The aerial device according to claim 5, further comprising wings connected to at least one of the body member and the retention member; wherein at least a portion of each of the wings is operable to rotatably move.

7. The aerial device according to claim 6, further comprising a travel propulsion member mounted on and extending from at least one of a portion of the body member and a portion of the retention member; wherein the travel propulsion member is operable to generate a travel thrust in a rearward direction.

8. The aerial device according to claim 5, further comprising: a controller in communication with the thruster and the plurality of fins; at least one sensor positionable in communication with the controller and operable to sense at least one characteristic data point; and a power unit in communication with the controller and the at least one sensor; wherein the at least one sensor is operable to provide a sensor signal based on the at least one characteristics data point; and wherein the controller is operable to selectively control at least one of the thruster and the fins based on the sensor signal.

9. The aerial device according to claim 8, further comprising: at least one body communication point carried by the body member; and at least one retention communication point carried by the retention member; wherein the controller and the power unit are carried by the body member and in communication with the at least one body communication point; wherein the at least one sensor is carried by the retention member and in communication with the at least one retention communication point; and wherein the at least one body communication point engages in communication with the at least one retention communication point when and while the rearward portion of the retention member is matingly engaged with the forward portion of the body member.

10. The aerial device according to claim 8, wherein the controller is operable to determine at least one path of travel based on the sensor signal; and wherein the controller is operable to selectively control at least one of the thruster and the fins based on the at least one path of travel determined.

11. An aerial device comprising: a body member; a retention member having an interior space and a rearward portion that is operable to matingly engage with a forward portion of the body member to carry a payload within the interior space in cooperation with the forward portion of the body member; at least one engagement member mounted on at least one of the body member and the retention member; a thruster to selectively provide a thrust force; and a plurality of fins connected to and extending from the body member; wherein the retention member includes a collar having an inner perimeter that defines an opening that is in communication with the interior space; wherein a forward portion of the payload extends through the opening while the rearward portion of the retention member is matingly engaged with the forward portion of the body member; wherein the inner perimeter of the collar substantially abuts a portion of an outer surface of the payload while the rearward portion of the retention member is matingly engaged with the forward portion of the body member; wherein the retention member includes at least one rail having an end portion connected to the collar; wherein at least a portion of an inward surface of the at least one rail significantly abuts a portion of the outer surface of the payload while the rearward portion of the retention member is matingly engaged with the forward portion of the body member; and wherein the at least one engagement member is operable to be reversibly engaged with an engagement unit of an aerial vehicle.

12. The aerial device according to claim 11, further comprising wings connected to at least one of the body member and the retention member; wherein at least a portion of each of the wings is operable to rotatably move.

13. The aerial device according to claim 12, further comprising a travel propulsion member mounted on and extending from at least one of a portion of the body member and a portion of the retention member; wherein the travel propulsion member is operable to generate a travel thrust in a rearward direction.

14. The aerial device according to claim 11, further comprising: a controller in communication with the thruster and the plurality of fins; at least one sensor positionable in communication with the controller and operable to sense at least one characteristic data point; and a power unit in communication with the controller and the at least one sensor; wherein the at least one sensor is operable to provide a sensor signal based on the at least one characteristics data point; wherein at least a portion of each one of the plurality of fins is operable to rotatably move; wherein the controller is operable to selectively control at least one of the thruster and the fins based on the sensor signal.

15. The aerial device according to claim 14, further comprising: at least one body communication point carried by the body member; and at least one retention communication point carried by the retention member; wherein the controller and the power unit are carried by the body member and in communication with the at least one body communication point; wherein the at least one sensor is carried by the retention member and in communication with the at least one retention communication point; and wherein the at least one body communication point engages in communication with the at least one retention communication point when and while the rearward portion of the retention member is matingly engaged with the forward portion of the body member.
16. The aerial device according to claim 14, wherein the controller is operable to determine at least one path of travel based on the sensor signal; and wherein the controller is operable to selectively control at least one of the thruster and the fins based on the at least one path of travel determined.
17. An aerial device comprising: a body member; a retention member comprising a collar and at least one rail that has a forward end portion connected to the collar; at least one engagement member mounted on at least one of the body member and the at least one rail; a thruster to selectively provide a thrust force; a plurality of fins connected to and extending from the body member; a controller in communication with the thruster and the plurality of fins; at least one sensor operable to sense at least one characteristic data point; a power unit in communication with the controller and positionable in communication with the at least one sensor; at least one body communication point carried by the body member; and at least one retention communication point carried by the at least one rail; wherein the collar and the at least one rail define an interior space; wherein the collar has an inner perimeter that defines an opening that is in communication with the interior space; wherein the at least one rail includes a rearward end portion that is operable to matingly engage with a forward portion of the body member to carry a payload within the interior space in cooperation with the forward portion of the body member; wherein a forward portion of the payload extends through the opening while the rearward portion of the at least one rail is matingly engaged with the forward portion of the body member; wherein both the inner perimeter of the collar and at least a portion of an inward surface of the at least one rail substantially abuts a portion of an outer surface of the payload while the rearward portion of the at least one rail is matingly engaged with the forward portion of the body member; wherein the at least one engagement member is operable to be reversibly engaged with an engagement unit of an aerial vehicle; wherein the at least one sensor is operable to provide a sensor signal based on the at least one characteristics data point; wherein at least a portion of each one of the plurality of fins is operable to rotatably move; wherein the controller is operable to selectively control at least one of the thruster and the fins based on the sensor signal; wherein the controller and the power unit are carried by the body member and in communication with the at least one body communication point; wherein the at least one sensor is carried by the retention member and in communication with the at least one retention communication point; and wherein the at least one body communication point engages in communication with the at least one retention communication point when and while the rearward portion of the at least one rail is matingly engaged with the forward portion of the body member.
18. The aerial device according to claim 17, further comprising wings connected to at least one of the body member and the at least one rail; wherein at least a portion of each of the wings is operable to rotatably move.
19. The aerial device according to claim 18, further comprising a travel propulsion member mounted on and extending from at least one of a portion of the body member and a portion of the at least one rail; wherein the travel propulsion member is operable to generate a travel thrust in a rearward direction.
20. The aerial device according to claim 17, wherein the controller is operable to determine at least one path of travel based on the sensor signal; and wherein the controller is operable to selectively control at least one of the thruster and the fins based on the at least one path of travel determined.